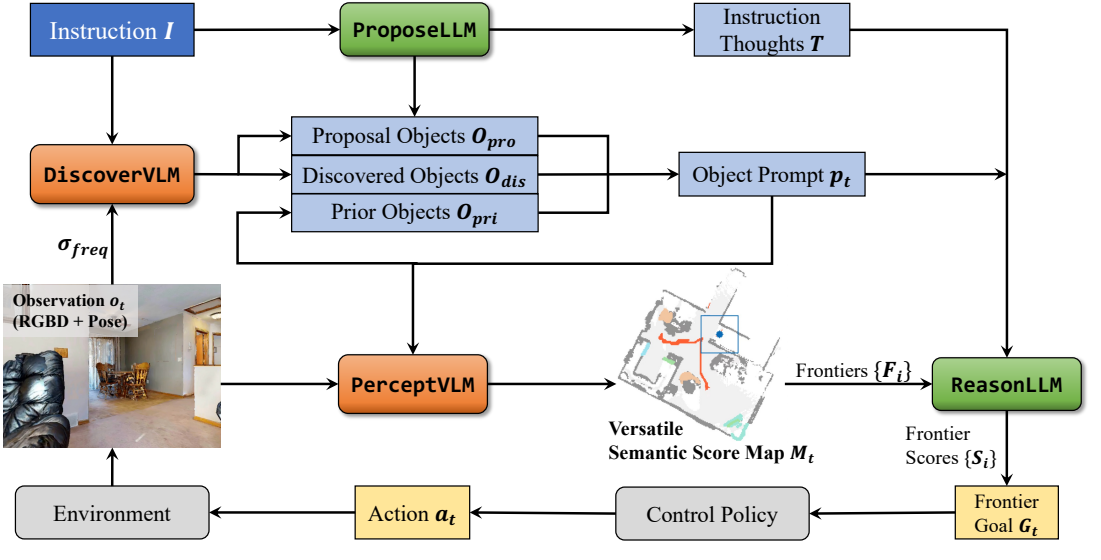

Instruction I
ProposeLLM
Instruction Thoughts T
DiscoverVLM
Proposal Objects O_{pro}
Discovered Objects O_{dis}
Prior Objects O_{pri}
Object Prompt p_t
σ_{freq}
Observation o_t (RGBD + Pose)
PerceptVLM
Versatile Semantic Score Map M_t
Frontiers $\{F_i\}$
ReasonLLM
Frontier Scores $\{S_i\}$
Environment
Action a_t
Control Policy
Frontier Goal G_t